

Cardio and Pharmacology

1. A 63 year old man with long standing hypertension has undergone multiple changes in his antihypertensive regimen over the years with only minimal improvement in his blood pressure. A CT scan of his abdominal region shows a small right kidney. Angiography demonstrates narrowing of the right renal artery due to atherosclerosis. Based off the angiogram study you determine that the flow in the right renal artery has decreased by a factor of 16 compared to the left. By what percentage has the radius of the lumen been reduced by?

- ☐ (A) 25%
- ☐ (B) 33%
- ☐ (C) 50%
- ☐ (D) 75%
- ☐ (E) 90%

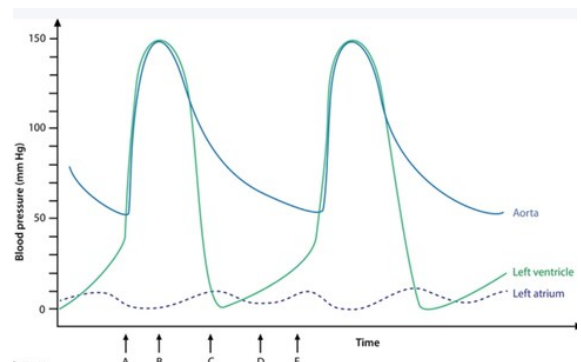
2. A 39 year old man comes to the office due to increasing fatigue for the past 2 weeks. The patient reports no prior medical conditions. Temperature is 37.2 C, BP is 116/66 mmHg, pulse is 100/min, and respirations are 20/min. Physical exam reveals crackles on auscultation of the lungs. Echocardiography reveals no valvular abnormalities and the following parameters: LVEDV: 125mL LVESV: 75mL Based on the observed findings, which of the following is the most accurate about this patient?

- ☐ (A) Cardiac Output is 4L/min
- ☐ (B) Cardiac Output is 6L/min
- ☐ (C) LV Ejection Fraction is 40%
- ☐ (D) LV Ejection Fraction is 60%
- ☐ (E) Stroke Volume is 40mL
- ☐ (F) Stroke Volume is 60mL

3. A 51 year old man is brought to the ER due to chest tightness that started 30 minutes prior to arrival. His chest discomfort is associated with shortness of breath and nausea. The patient was shoveling snow off his driveway when his symptoms began. He has a history of hypertension and type 2 diabetes mellitus. ECG reveals ST elevation in leads representing the lateral wall of the left ventricle. Troponin, CK-MB and myoglobin levels are elevated. Emergent cardiac catheterization in this patient will most likely show occlusion of which of the following arteries?

- (A) Left Anterior Descending Artery
- (B) Posterior Descending Artery
- (C) Nodal Arteries
- (D) Left Circumflex Artery
- (E) Diagonal Artery

4. A 52 year old man comes to the office due to a 2 week history of progressive fatigue and exertional dyspnea. The patient is a postal delivery worker and reports that he gets short of breath after walking a few blocks. He has not had any chest pain or palpitations. The patient does not use tobacco, alcohol or illicit drugs. Cardiac auscultation reveals a decrescendo diastolic murmur which is best heard when leaning forward. Recent cardiac catheterization reveals the graph below. This patient's murmur would most likely be loudest at which time point on the graph?

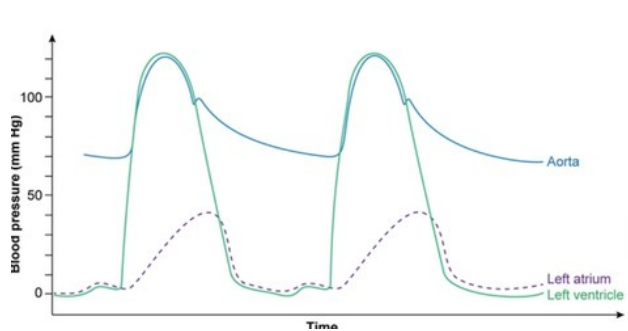


- (A) A
- (B) B
- (C) C
- (D) D
- (E) E

5. A 10 year old boy is brought to the clinic by his parents due to chronic fatigue, shortness of breath and failure to gain weight. The child was recently adopted from an orphanage in Asia. He weighs 20kg. Vital signs are normal. Physical exam shows a loud systolic murmur. Echo shows pulmonary stenosis and subaortic ventricular septal defect with deviation of the origin of the aorta to the right. The mother describes occasional episodes of severe dyspnea and cyanosis during which the child quickly assumes sitting position. Which of the following mechanisms during squatting relieve this child's symptoms?

- (A) Decreasing left ventricular workload
- (B) Decreasing pulmonary blood flow
- (C) Decreasing pulmonary vascular resistance
- (D) Improving lung compliance
- (E) Increasing systemic vascular resistance

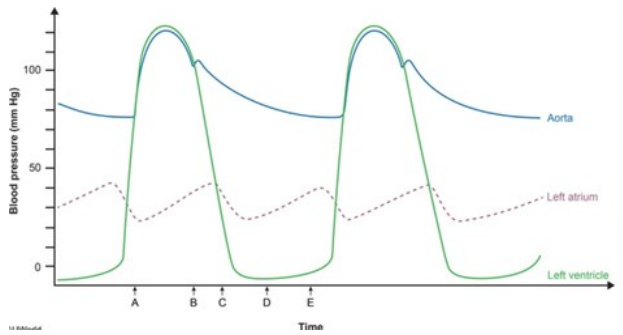
6.



A 54 year old woman comes to the office due to exertional dyspnea and fatigue. The patient gets short of breath while walking to a store 2 blocks away from her home. She also has trouble breathing at night and sleeps with 2 or 3 pillows to prop her up. She has had occasional palpitations but no chest pain. The patient takes no medications on a regular basis. After initial evaluation, cardiac exam reveals a holosystolic murmur and cardiac catheterization reveals the graph below. Which of the following is the most likely diagnosis?

- (A) Aortic Regurgitation
- (B) Aortic Stenosis
- (C) Mitral Stenosis
- (D) Mitral Regurgitation
- (E) Mitral Valve Prolapse

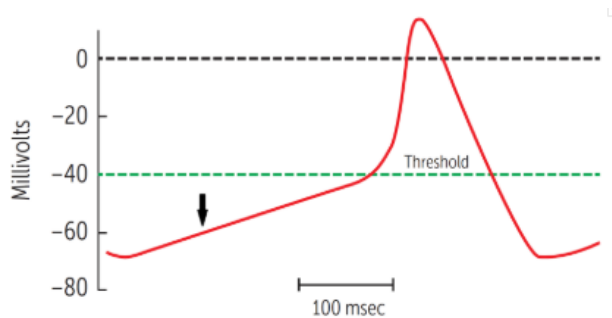
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A 32 year old woman comes to the office due to progressive exertional dyspnea and easy fatigability. She immigrated to the United States from India several years ago and reports a history of rheumatic fever during childhood. Cardiac auscultation reveals a diastolic high frequency sound and rumbling diastolic murmur heard best in the left fifth intercostal space at the midclavicular line. Cardiac catheterization is performed and results are represented below. Timing of the high frequency sound heard during cardiac auscultation best corresponds to which of the following letters?

- (A) A
- (B) C
- (C) D
- (D) E
- (E) B

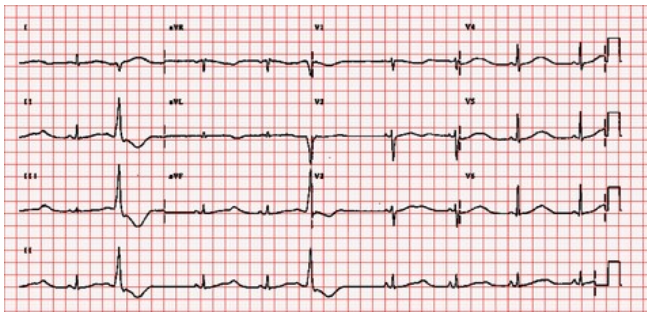
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An experimental study is designed to characterize the conductance of ions across myocytes located in the sinoatrial node. An experimental animal is given a dose of norepinephrine, resulting in an increase in its heart rate. An electrophysiologic graph is mapped across the heart, and the findings from the sinoatrial nodal area are shown. Which of the following ion/ion channels is responsible for the section of the action potential indicated by the arrow?

- (A) Decreased conductance of Cl^-
- (B) Decreased conductance of HCO_3^-
- (C) Decreased conductance of K^+
- (D) Increased conductance of Ca^{2+}
- (E) Increased conductance of Na^+

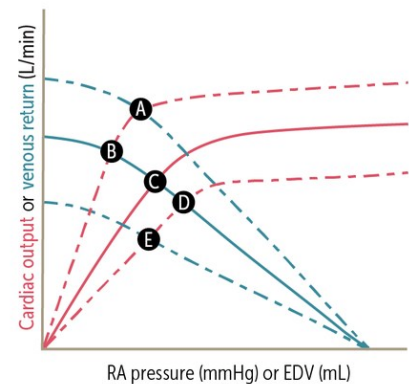
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A 12-year-old boy who has had sensorineural deafness since birth is brought to the emergency department by his parents 1 hour after their son felt lightheaded and fainted after running around the park. He takes no medications. Family history includes a maternal uncle who died suddenly in childhood. On physical examination, blood pressure is 120/70, pulse is 108/min, and respiratory rate is 18/min. Oxygen saturation is 98% on room air. No other abnormalities are noted. Laboratory values are within reference range. Results of electrocardiography are shown. Which of the following is the most common mutation associated with this patient's condition?

- (A) Funny Na^+ channels
- (B) L-Type Ca^{2+} channels
- (C) Ryanodine receptors
- (D) Voltage gated potassium channels
- (E) Voltage gated sodium channels

10. A 54-year-old man comes to the clinic with fatigue, shortness of breath, and an uncomfortable feeling of a racing heartbeat. He has a history of poorly controlled hypertension. ECG tracing reveals an irregularly irregular heartbeat. The physician prescribes a drug that inhibits myocardial sodium potassium adenosine triphosphatase. Which of the following points best depicts the change following administration of the prescribed drug (solid lines indicate normal)?



- (A) B
- (B) C
- (C) A
- (D) E
- (E) D

11.

Choice	Peripheral vascular resistance	Cardiac output	Pulmonary capillary wedge pressure
A	↓	↓	↓
B	↓	↑	↓
C	↑	↓	↓
D	↑	↓	↑
E	↑	↑	↑

A 91-year-old man is brought to the emergency department by his daughter after being found unresponsive in his home. The patient's oral temperature is 39.8° C (103.6° F), blood pressure is 80/50 mm Hg, pulse is 120/min, and respiratory rate is 26/min. Initial laboratory tests reveal leukocytosis and an elevated lactate level. The patient's blood pressure remains at 80/50 mm Hg despite adequate hydration. Which of the following is the most likely hemodynamic pattern in this patient?

- (A) C
- (B) A
- (C) D
- (D) E
- (E) B

12. A 45-year-old obese man who recently emigrated from India comes to the clinic because of shortness of breath and hoarseness. He reports that the hoarseness started within the past week. His medical history is significant for a childhood illness that he is unable to name. Physical examination is normal apart from a body mass index of 33. An electrocardiogram shows enlarged biphasic P waves in lead II, and cardiac echocardiography shows a stenosed mitral valve with left atrial enlargement. His hoarseness is believed to be caused by a nerve impingement. The impinged nerve is a branch of a larger nerve that controls which of the following muscles?

- (A) Palatoglossus
- (B) Genioglossus
- (C) Mylohyoid
- (D) Styloglossus
- (E) Stylopharyngeus

13. A 13 year old female presents to the pediatrician for a physical exam. On inspection she has a webbed neck and is short for her age. On physical exam she has a blood pressure of 120/70 mmhg in her upper limbs and a blood pressure of 90/50 mmhg in her lower limbs. What other defect would the patient present with?

- ☐ (A) Epicanthal Folds
- ☐ (B) Rib Notching
- ☐ (C) Cardiomegaly
- ☐ (D) VSD
- ☐ (E) Ectopic Cordis

14. A 75 y/o female presents to the hospital with complaints of shortness of breath, and feeling faint. On physical exam her vital signs are 100bpm, 90/60 mmhg, 90% o2 saturation and 98.0 f. On auscultation a crescendo-decrescendo murmur is heard. What would explain the reasoning behind her symptoms?

- ☐ (A) Increased Stroke Volume
- ☐ (B) Decreased Stroke Volume
- ☐ (C) Decreased Ventricular Pressures
- ☐ (D) Decreased EDV
- ☐ (E) Increased ESV

15.

	Pulse Pressure	Stroke Volume	Systolic Pressure
A	↑	↓	↓
B	↑	↑	↑
C	↑	↓	↓
D	↓	↑	↑
E	↓	↑	↓
F	↓	↓	↑

A 50 y/o female visits her family practitioner for a regular visit. Her vital signs show a bp of 160/70 mmhg and a pulse of 74 bpm. On auscultation a diastolic murmur is heard and she is diagnosed with moderate aortic regurgitation. What changes would be expected in this patient?

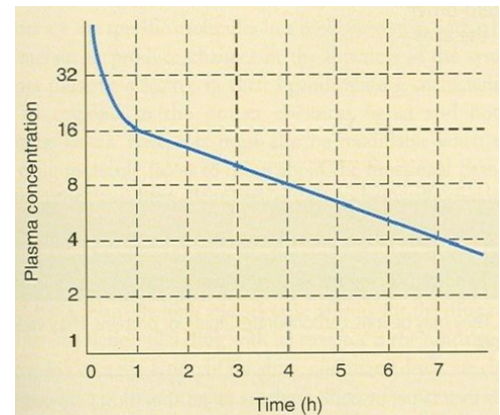
- ☐ (A) C
- ☐ (B) A
- ☐ (C) D
- ☐ (D) E
- ☐ (E) B

- 16.** A 3 year old boy was rushed to the ER with mental confusion, restlessness, incoherence, and hallucinatory behavior. His mother states that the child has eaten several black berries of a wild plant while playing with friends in the woods. Physical examination revealed mydriasis, dry hot and scarlet skin, and distended abdomen with no bowel sounds. Vital signs were temperature of 104.5 F, Heart rate 145 bpm, BP 105/60 mmHg. Which of the following drugs would antagonize most of the symptoms exhibited by the patient?
- ☐ (A) Neostigmine
 - ☐ (B) Albuterol
 - ☐ (C) Physostigmine
 - ☐ (D) Epinephrine
 - ☐ (E) Prazosin
- 17.** A 16 year old male snorted cocaine for the first time at a party. A few minutes later he felt euphoric, and friends noted that his pupils were dilated. Which of the following molecular actions most likely mediated the mydriasis?
- ☐ (A) Inhibition of norepinephrine reuptake into sympathetic terminals
 - ☐ (B) Stimulation of epinephrine release from adrenal medulla
 - ☐ (C) Stimulation of norepinephrine release from sympathetic terminals
 - ☐ (D) Blockade of alpha 2 receptors in sympathetic terminals
 - ☐ (E) Inhibition of MAO A
- 18.** A 63-year-old woman is given an IV infusion of atropine. The half-life of atropine is 2 h. The infusion is begun at 7 am. At 1 pm a blood sample is taken and the atropine plasma concentration is found to be 4 mg/L. Which of the following will the atropine plasma concentration be at steady-state
- ☐ (A) 4.6 mg/L
 - ☐ (B) 5.1 mg/L
 - ☐ (C) 4.2 mg/L
 - ☐ (D) 6.5 mg/L
 - ☐ (E) 3.9 mg/L

19. A 54 y/o woman is rushed to the ER following a stroke like episode. She presents with paralysis on the left side of her body and lack of facial muscle stimulation on the right. When asked about what medications she takes, she states that she had taken Aspirin that morning due to a headache. CT scans shows evidence of a stroke affecting her right hemisphere. What characteristics does aspirin show on a concentration vs time plot graph?

- ☐ (A) Are more common than those showing first order kinetics
- ☐ (B) Decrease in concentration exponentially with time
- ☐ (C) Have a half life independent of dose
- ☐ (D) Shows a plot of drug concentration vs. time that is linear
- ☐ (E) Shows a constant fraction of the drug eliminated per unit of time

20. A new drug was administered intravenously, and its plasma levels were measured for several hours. A graph was prepared as shown below, with the plasma levels plotted on a logarithmic ordinate and time on a linear abscissa. It was concluded that the drug has first order kinetics. From this graph, what is the best estimate of the half-life?



- ☐ (A) 0.5 h
- ☐ (B) 1 h
- ☐ (C) 3 h
- ☐ (D) 4 h
- ☐ (E) 7 h

21. You administer a 100 mg tablet of drug X to a patient every 24 hours and achieve an average steady-state plasma concentration of the drug of 10 mg/L. If you change the dose regimen to one 50 mg tablet every 12 hours, what will be the resulting average plasma concentration (in mg/L) of the drug after five half-lives?

- ☐ (A) 2.5
- ☐ (B) 5
- ☐ (C) 10
- ☐ (D) 20
- ☐ (E) 40

- 22.** A patient with a bacterial infection requires intravenous antibiotic therapy. The chosen drug has a clearance of 70 mL/min. the apparent volume of distribution is 50 L. The plan is to administer the drug intravenously every 6 hours and achieve a 4 mg/L steady-state blood level of the drug. No loading dose strategy is to be used. What maintenance dose is needed achieve this?
- (A) 14 mg
 - (B) 24 mg
 - (C) 300 mg
 - (D) 1200 mg
 - (E) 100 mg
- 23.** A 55 y/o woman with heart failure is to be treated with a diuretic drug. Drugs X and Y have the same mechanism of diuretic action. Drug X in a dose of 5 mg produces the same magnitude of diuresis as 500 mg of Drug Y. Which of the following statements best describes these results?
- (A) Drug Y is less efficacious than Drug X
 - (B) Drug X is about 100 times more potent than Drug Y
 - (C) Toxicity of Drug X is less than that of Drug Y
 - (D) Drug X has a wider therapeutic window than Drug Y
 - (E) Drug X will have a shorter duration of action than drug Y because less of drug X is present over the time course of drug action
- 24.** We administer an acidic drug (A) with a pKa of 3.4 orally. But pH is 1.4, and plasma pH is 7.4. Assume the drug crosses membranes by simple passive diffusion (e.g: no transporters are involved). Which of the following observations would be true?
- (A) Only ionized forms of the Drug A⁻ will be absorbed from the gut into the plasma
 - (B) The concentration ratio of total drug A⁻: HA would be 10000:1
 - (C) The drug will be hydrolyzed by a reaction with HCl and so cannot be absorbed
 - (D) The drug will not be absorbed unless we raise gastric pH to equal pKa as might be done with an antacid
 - (E) The drug would be absorbed and at equilibrium the plasma concentration of the nonionized moiety (HA) would be 10⁴ times higher than the plasma concentration of A⁻

- 25.** A 72 year old man suffering from prostatic hyperplasia complained to his physician that he had increased difficulty and pain in voiding his bladder. The patient had been taking an over the counter nasal decongestant for 3 days because of an annoying common cold. Which of the following drugs most likely contributed to his dysuria?
- ☐ (A) Albuterol
 - ☐ (B) Phenylephrine
 - ☐ (C) Clonidine
 - ☐ (D) Norepinephrine
 - ☐ (E) Dopamine
- 26.** A 25 year old comes into the ER complaining of xerostomia after a radiotherapy session at her oncologist office to treat her cervical lymphoma. At the office she is given a medication that helps treat her xerostomia and is also known to treat glaucoma. What pathway intermediates are expected to be elevated?
- ☐ (A) Increased Adenyl Cyclase
 - ☐ (B) Decreased Na^+
 - ☐ (C) Increased Na^+
 - ☐ (D) Increased Ca^{2+} and IP3/DAG
 - ☐ (E) Decreased Ca^{2+} and IP3/DAG
- 27.** A 15 year old boy comes into the ER with ptosis, diplopia, facial weakness and proximal limb weakness. His mother states that his symptoms get worse throughout the day and are better early in the morning. What drug would be a good diagnostic indicator for this disease?
- ☐ (A) Pilocarpine
 - ☐ (B) Sarin
 - ☐ (C) Neostigmine
 - ☐ (D) Pyridostigmine
 - ☐ (E) Edrophonium
- 28.** A 40 year old woman has just underwent surgery and complains to the physician that she has yet to urinate since the surgery and it has been 4 hours since. What drug should the physician administer in order to promote urination?
- ☐ (A) Hexamethonium
 - ☐ (B) Atropine
 - ☐ (C) Nicotine
 - ☐ (D) Bethanechol
 - ☐ (E) Acetylcholine

- 29.** A 25 year old healthy male volunteer received an IV infusion of a new drug in a phase I clinical trial. It was found that the drug causes an increase in cardiac contractility, a slight increase in heart rate, and no major changes or a decrease in systemic vascular resistance. Which of the following drugs does the new agent most resemble?
- ☐ (A) Phenylephrine
 - ☐ (B) Norepinephrine
 - ☐ (C) Isoproterenol
 - ☐ (D) Epinephrine
 - ☐ (E) Dobutamine
- 30.** A 45-year-old man visits the clinic for an annual physical. He has a history of hypertension, and his medications include hydrochlorothiazide and metoprolol. At the clinic, his blood pressure reading is 168/95 mm Hg. On a follow-up visit 3 weeks later, his blood pressure remains elevated at 160/90 mm Hg. Physical examination reveals a rotund man with androgenic baldness. Normal S1 and S2 heart sounds are heard. An early diastolic murmur is heard in the fifth intercostal space at the midclavicular line. He is subsequently prescribed a centrally acting medication to provide additional control of his hypertension. The patient is counseled about the importance of being compliant with the new medication. Suddenly stopping this drug can lead to a severe increase in blood pressure. Which of the following describes the mechanism of action of this medication?
- ☐ (A) Alpha-2 receptor agonism
 - ☐ (B) Inhibition of ACE
 - ☐ (C) Alpha 1 receptor agonism
 - ☐ (D) Alpha-2 receptor antagonism
 - ☐ (E) Beta-1 receptor antagonism